

Basil C K

AI Engineer Intern | AI & Machine Learning | Computer Vision

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Kochi, Kerala, India

Profile

AI Engineer Intern with hands-on experience developing production-oriented AI applications, Python backend systems, REST APIs, LLM integrations, and computer vision solutions. Skilled in building end-to-end AI pipelines from data processing and model development to deployment and API integration. Experienced with FastAPI, Computer Vision, Generative AI, and automation systems. Passionate about building scalable AI products that solve real-world business problems.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

JCT College of Engineering and Technology, Anna University

2022 – 2026

CGPA: 8.47 / 10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Machine Learning

TECHNICAL SKILLS

Programming & Backend:

Python (automation, backend logic, ML pipelines), REST APIs, FastAPI, Flask, API Development

Machine Learning & Computer Vision:

CNN architectures, YOLOv8 object detection, OpenCV image processing, Computer Vision, model training & evaluation, NLP fundamentals, Generative AI concepts

Data & ML Foundations:

Pandas, NumPy, supervised learning fundamentals, SQL (queries, joins, basic schema design)

Systems & Networking:

Operating Systems fundamentals, HTTP/ REST architecture, client-server systems, TCP/ IP basics

Cloud & Platforms:

Basic exposure to OCI / AWS fundamentals

Tools & Development:

Git (branching, commits, workflows), Jupyter Notebook, VS Code

AI & LLM Exposure:

LLM Integration, Chatbot Development, Prompt Engineering, conversational AI workflows

Exposure to:

Linux (CLI, permissions, processes), Docker (containerization basics), CI/ CD pipelines (conceptual understanding), ML model deployment workflows

EXPERIENCE

AI Engineer Intern

06/2026 – Present

Difinity Digital

Kochi, Kerala

- Developing AI-powered business applications using Python, FastAPI, REST APIs, and modern AI frameworks.
- Working on real-world client projects involving Generative AI, Computer Vision, and workflow automation.
- Designing backend services and API integrations for AI-driven systems.
- Participating in solution architecture, debugging, testing, and deployment workflows.
- Collaborating with cross-functional teams to transform business requirements into AI-enabled products.
- Contributing to scalable and maintainable production code following software engineering best practices.

PROJECT EXPERIENCE

Hybrid CNN-YOLO Car Damage Detection [🔗](#)

Tech Stack: Python, YOLOv8, TensorFlow/Keras, PyTorch, FastAPI, OpenCV, HTML/CSS

- Built a computer vision pipeline combining YOLOv8 detection and CNN classification for vehicle damage analysis
- Designed multi-stage inference system for damage localization and severity prediction
- Achieved mAP@0.5 of 0.659 and validation accuracy of 54.4% on custom dataset
- Developed a FastAPI-based interface for real-time image upload and prediction

Vexilon – AI-Powered Chrome Assistant [🔗](#)

Tech Stack: Python, Flask, REST APIs, LLM APIs, Chrome Extension, Git

- Built a Chrome extension integrated with a Flask backend enabling real-time AI-based conversational interaction
- Designed RESTful APIs for request handling and response streaming
- Integrated external LLM APIs (Gemma, Ollama) using OpenAI-compatible formats
- Implemented modular backend structure with session-based interaction workflows

Vocalix – AI Voice Assistant [🔗](#)

Tech Stack: Python, Speech Recognition, LLM APIs, Automation Libraries

- Developed a voice-controlled assistant for executing system-level tasks
- Integrated LLM APIs for conversational responses
- Implemented speech-to-command processing and automation workflows
- Designed structured interaction pipeline for real-time voice-based control

ADDITIONAL LEARNING & CERTIFICATIONS

- **IBM Data Science** – Selected Coursework (Python, Data Analysis)
- **Oracle Cloud Infrastructure 2023** – AI Foundations Associate
- **Google AI Essentials**

LANGUAGES

English | Malayalam | Tamil